

2.1 Analyze algorithms for correctness

الگورتھم تب درست (Correct) سمجھا جاتا ہے جب دیے گئے ان پٹ پر اس کا آؤٹ پٹ وہی ہو جو ہونا چاہیے۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 01 of 24

2.1 Analyze algorithms for correctness

LEARN IT

An algorithm is **correct** if its **output matches the expected results** for a given input. Analysing correctness means showing that the algorithm produces the intended output for **all valid inputs**, not just one lucky example.

Two methods are used in this course:

Trace tables



paper execution, line by line



Stepwise reasoning



logic blocks, overall state

You already met the idea of a dry run in Class XI. Here you learn to do it formally, and to choose the method that fits the size of the algorithm.

WORKED EXAMPLE

Worked example: what “correct” means. An algorithm is supposed to count how many of the heights 1.91, 2.01, 1.87, 1.57, 1.48 lie between 1.57 m and 1.91 m inclusive.

Expected output: **3** (1.91, 1.87 and 1.57 qualify; 2.01 is too tall; 1.48 is too short).

If a program prints 4, it **runs** but it is **not correct**. Correctness is about the right answer, not about “no crash”.

REMEMBER IT

Remember: **runs ≠ correct**. A logical error still produces output — just the wrong output. That is why we analyse correctness on paper before we trust the code.

COMMON MISTAKE

Mistake: testing only one “normal” input and calling the algorithm correct.

Fix: also try edge cases (empty list, $N = 0$, the exact boundary values 1.57 and 1.91). The definition says **all valid inputs**.

اردو میں سمجھیں

الگورتھم تب درست (Correct) سمجھا جاتا ہے جب دیے گئے ان پٹ پر اس کا آؤٹ پٹ وہی ہو جو ہونا چاہیے۔ صرف چل جانا کافی نہیں؛ منطقی غلطی سے غلط جواب بھی نکل سکتا ہے۔ درستگی جانچنے کے دو طریقے ہیں: ٹریس ٹیبل (Trace Table) اور مرحلہ وار استدلال (Stepwise Reasoning)۔

CHECK YOURSELF

1. When is an algorithm considered correct?
2. Name the two methods used to analyse correctness in this chapter.
3. Why is “the program ran without an error” not enough?

Answers: 1. When its output matches the expected results for a given input (and does so for all valid inputs). 2. Trace tables and stepwise reasoning. 3. Because a logical error can still produce output — the wrong output.

Next lecture: 2.1.1 Trace Tables